

PURUSHOTHAMAN K.S.

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SUMMARY

Ambitious Computer Science Engineering student specializing in full-stack web development, Artificial Intelligence, and "Zero-UI Sovereignty" architectures. Experienced in building scalable applications and integrating local LLMs using Python, Next.js, FastAPI, and Firebase. Adept at managing server infrastructure and rapid problem-solving to deliver high-performance, secure solutions.

EDUCATION

B.E. in Computer Science Engineering (Final-Year Student)

R.P. Sarathy Institute of Technology (RPSIT) | Salem, Tamil Nadu

•Batch: 2023-2027 | CGPA: 7.04/10.0

TECHNICAL SKILLS

- **Programming Languages:** Python, Java & JavaScript
- **Frontend Technologies:** HTML & CSS, Tailwind CSS.
- **Backend & Cloud:** FastAPI, Firebase, AWS Lightsail
- **AI & Machine Learning:** Python for AI/ML, Ollama, Local LLM Integration, Voice Recognition
- **DevOps & Networking:** Computer Networks, Server Management
- **Operating Systems & Tools:** Linux (Arch/Garuda), Hyprland Window Manager

EXPERIENCE

Full-Stack Development Intern

Rootsotech | Dharmapuri, Tamil Nadu

- Completed an intensive 5-day internship focused on full-stack web development practices.
- Gained hands-on experience with modern web technologies (React, Node.js) to improve software development workflows.

Intern

Imagecon | Salem, Tamil Nadu

- Completed a 15-day internship, developing a secure and scalable patient management system.
- Utilized the Django web framework to streamline data handling and implemented security features such as user authentication and data encryption.

PROJECTS

SystemAssist | FastAPI, Three.js

- Developed a real-time system monitoring platform to track and visualize system metrics (e.g., CPU usage, RAM allocation, network traffic).
- Utilized FastAPI for high-performance backend data delivery and Three.js for interactive, 3D data visualization on the frontend.

OpenWebUI AI Platform | Llama 3, AWS EC2

- Deployed a high-performance, private AI platform powered by Llama 3 on an AWS EC2 instance, featuring a minimalist Obsidian Pro-inspired user interface.
- Implemented a "Zero-UI Sovereignty" architecture that processes administrative tasks server-side to ensure robust data security and privacy.

All-in-One AI Platform | Firebase

- Developed and successfully deployed a centralized platform offering various AI tools and functionalities (e.g., text generation, image analysis).
- Utilized Firebase for robust hosting and backend services.

CERTIFICATIONS & COURSES

- **Software Engineering (e-Certificate):** Successfully completed a comprehensive 12-week NPTEL examination and course.
- **Flutter & Firebase Workshop:** Participated in hands-on technical training at Sona College of Technology (March 2026).
- **Prompt-A-Thon (ZEN-IT-TRIX 1.0):** Competed in the technical symposium event at Annapoorna Engineering College (March 2026).
- **Threads'25 Technical Symposium:** Registered and participated in the CSE department event at Sona College of Technology.